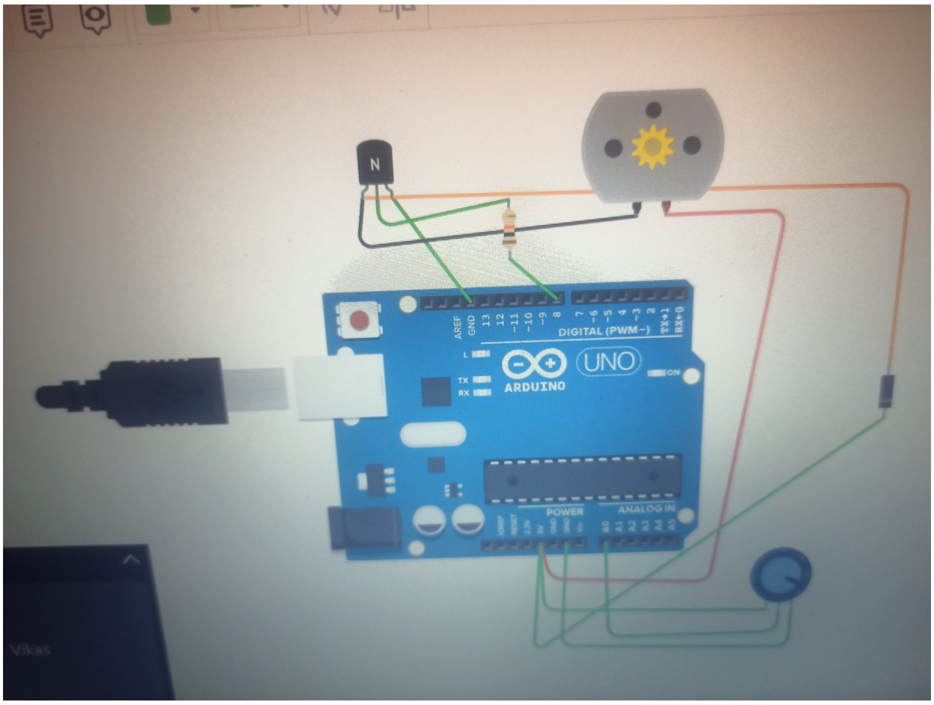
AUTOMATIC SOIL MOISTURE IRRIGATION SYSTEM

Arduino UNO Based Automatic Plant Watering System



Project concept: The system uses a potentiometer as a simulated soil-moisture sensor. When the sensor value falls below the threshold, Arduino switches ON the motor/pump automatically.

1. Project Name

Automatic Soil Moisture Based Irrigation System Using Arduino UNO

This project is an automatic irrigation system designed to water plants according to soil moisture conditions. In the uploaded Tinkercad circuit, a potentiometer is used as a soil-moisture sensor simulator. Arduino UNO continuously reads the analog value from A0. If the value becomes lower than the programmed threshold of 500, the Arduino activates the motor/pump through a transistor switching stage. When the value reaches or exceeds 500, the motor is switched OFF. Thus, watering is performed automatically without continuously operating the pump.

2. Objective

• To design an automatic irrigation system using Arduino UNO.

• To monitor soil moisture continuously using an analog sensor input.

• To automatically start the water pump when the soil becomes dry.

• To automatically stop the pump when sufficient moisture is detected.

• To reduce unnecessary water usage by watering only when required.

• To demonstrate sensor-based automatic control using Arduino.

3. Components Required

Component	Purpose
Arduino UNO	Main controller that reads the sensor and controls the pump.
Potentiometer	Used in the Tinkercad model as a simulated soil-moisture sensor.
DC Motor / Water Pump	Represents the irrigation pump that supplies water.
NPN Transistor	Acts as an electronic switch for controlling the motor.
Resistor	Limits the base/gate control current and protects the Arduino output.
Diode	Provides protection against voltage spikes generated by the motor.
Jumper wires	Provide electrical connections between components.
Power supply	Provides the required power for Arduino and motor circuit.

4. Circuit Description and Connections

The potentiometer's wiper is connected to Arduino analog pin A0. It provides a variable analog value from approximately 0 to 1023, which represents different moisture conditions in the simulation. Digital pin D8 controls the transistor switching stage. The transistor then switches the DC motor/pump ON or OFF. A diode is placed across the motor to suppress the motor's back-EMF. The Arduino and motor circuit should have a common ground.

Connection	Arduino / Circuit	Purpose
Sensor output	A0	Reads simulated soil-moisture value
Motor control	D8	Turns pump switching stage ON/OFF
Sensor supply	5V and GND	Powers potentiometer/sensor circuit

Connection	Arduino / Circuit	Purpose
Motor supply	External/motor supply as appropriate	Supplies motor power
Diode	Across motor terminals	Suppresses reverse voltage spike

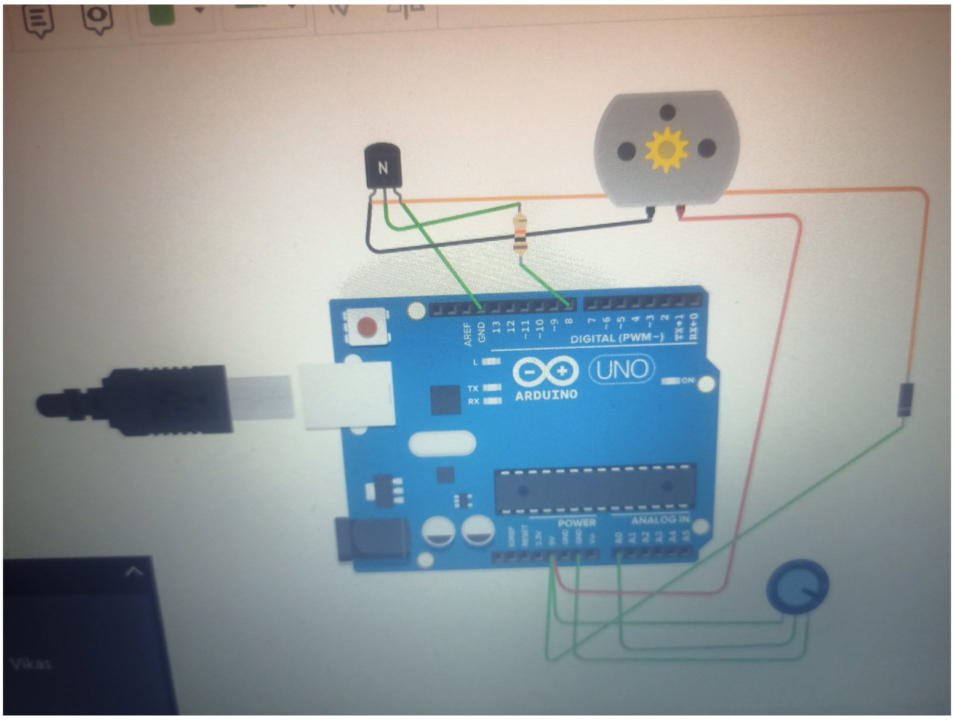


Figure 1: Uploaded automatic irrigation circuit.

5. Working Principle

1. Moisture sensing: The potentiometer simulates the output of a soil-moisture sensor. Its position changes the analog reading.

2. Analog reading: Arduino reads the sensor value through A0 using its 10-bit ADC, producing a value between 0 and 1023.

3. Threshold comparison: The program compares the sensor value with the threshold value of 500.

4. Dry soil condition: If sensorValue is less than 500, the program sets D8 HIGH and the pump is switched ON.

5. Wet soil condition: If sensorValue is 500 or higher, D8 is set LOW and the pump is switched OFF.

6. Continuous monitoring: The loop repeats every 500 ms, allowing the system to respond automatically to changes in the sensor value.

6. Arduino Program

```
// Automatic Irrigation System
// Potentiometer as Soil Moisture Sensor

const int sensorPin = A0;
const int motorPin = 8;
const int threshold = 500;

void setup() {
pinMode(motorPin, OUTPUT);
Serial.begin(9600);
}
```

```
void loop() {
int sensorValue = analogRead(sensorPin);

Serial.print("Sensor Value: ");
Serial.println(sensorValue);

if (sensorValue < threshold) {
digitalWrite(motorPin, HIGH); // Pump ON
Serial.println("Soil Dry - Pump ON");
} else {
digitalWrite(motorPin, LOW); // Pump OFF
Serial.println("Soil Wet - Pump OFF");
}

delay(500);
}
```

Uploaded program screenshot:

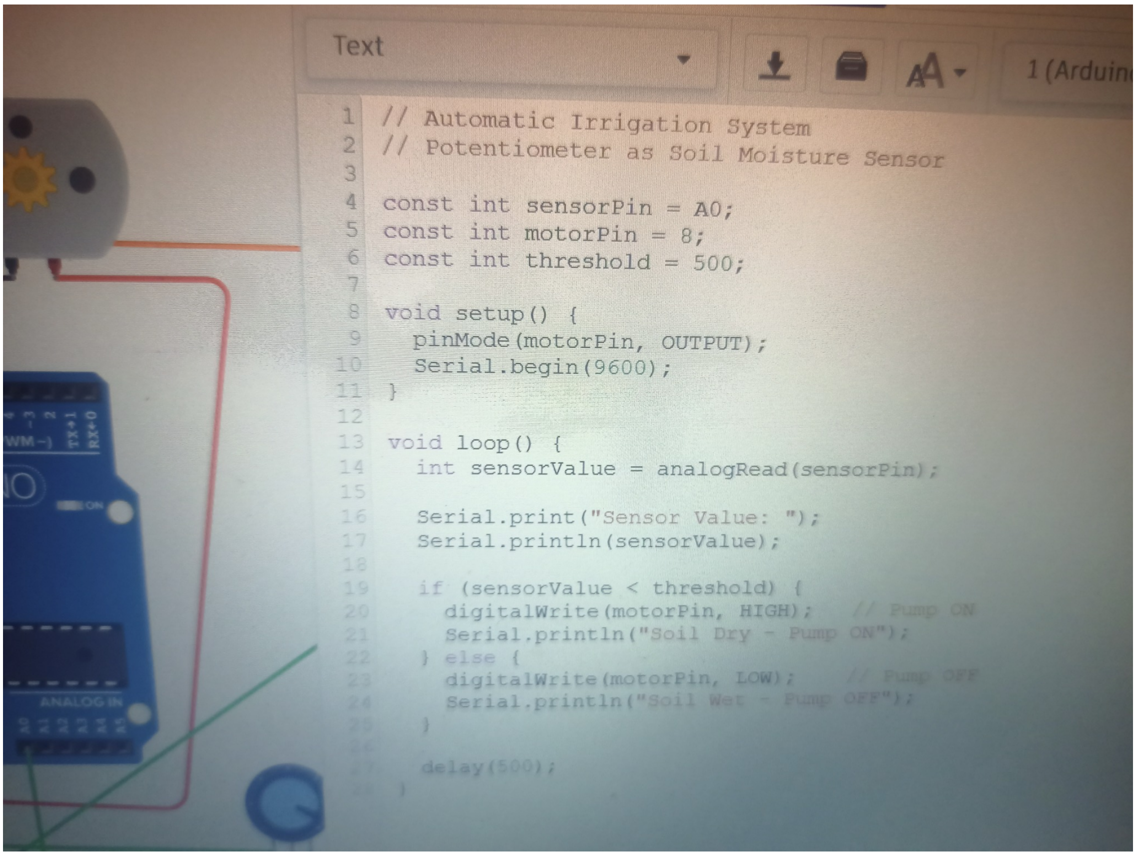


Figure 2: Uploaded Arduino program.

7. Program Explanation

• **sensorPin = A0:** selects A0 for reading the analog sensor.

• **motorPin = 8:** assigns digital pin 8 to the motor control circuit.

• **threshold = 500:** sets the moisture decision point.

• **pinMode(motorPin, OUTPUT):** makes D8 an output.

• **Serial.begin(9600):** starts communication with the Serial Monitor.

• **analogRead(sensorPin):** reads the current sensor value.

• **if (sensorValue < threshold):** treats a lower reading as the dry-soil condition and starts the pump.

• **digitalWrite(HIGH):** sends the ON control signal to the switching stage.

• **else / LOW:** stops the pump when the sensor value is at or above the threshold.

• **delay(500):** waits half a second before taking the next reading.

8. Algorithm

• Start the system.

• Initialize motor pin D8 as OUTPUT and Serial Monitor at 9600 baud.

• Read the soil-moisture sensor value from A0.

• Compare the value with 500.

• If value < 500 → soil is considered dry → pump ON.

• If value ≥ 500 → soil is considered sufficiently wet → pump OFF.

• Display sensor value and pump status on Serial Monitor.

• Wait 500 ms and repeat.

9. Applications

• **Home gardening:** Automatically waters household plants when the soil becomes dry.

• **Agricultural irrigation:** Can be developed into an automatic field irrigation controller.

• **Greenhouses:** Helps maintain suitable soil moisture for plants.

• **Nurseries:** Reduces the need for manual watering of many plants.

• **Smart farming:** Can be integrated with IoT systems for remote monitoring and control.

• **Water conservation:** Helps avoid unnecessary watering by operating the pump only when required.

10. Advantages

• Automatic operation with minimum human intervention.

• Can reduce water wastage.

• Simple and inexpensive prototype.

• Easy to modify the threshold according to plant requirements.

• Can be expanded with LCD, sensors, relay modules and IoT connectivity.

• Suitable for educational and practical embedded-system projects.

11. Expected Output / Result

Sensor condition	Arduino D8	Pump/Motor	Result
Sensor value < 500	HIGH	ON	Soil considered dry; irrigation starts.
Sensor value ≥ 500	LOW	OFF	Soil considered wet; irrigation stops.

12. Limitations

• In the Tinkercad simulation, the potentiometer represents soil moisture; a real project should use a suitable soil-moisture sensor.

• The threshold value 500 is an example and should be calibrated for the actual sensor and soil.

• A DC motor/pump should not normally be powered directly from an Arduino I/O pin; use a transistor/MOSFET or relay driver and an appropriate external supply.

• The basic system does not measure water level, flow rate or rainfall.

• The current design provides local automatic control and Serial Monitor information, but no mobile notification.

13. Future Scope

• Replace the potentiometer with a real capacitive soil-moisture sensor.

• Add a water-level sensor to prevent dry running of the pump.

• Add an LCD/OLED to show moisture level and pump status.

• Add ESP8266/ESP32 for mobile notifications and IoT monitoring.

• Use multiple moisture thresholds for different plants.

• Add rain detection to prevent irrigation during rainfall.

• Record moisture readings for analysis and smart irrigation scheduling.

14. Conclusion

The Automatic Soil Moisture Irrigation System is a simple and useful Arduino project that demonstrates automatic control based on sensor input. The potentiometer in the Tinkercad model simulates soil moisture and provides an analog value to A0. Arduino compares this value with the threshold of 500. When the value is below 500, the system considers the soil dry and activates the motor/pump through the switching circuit. When the value is 500 or above, the pump is turned OFF. The project can therefore reduce unnecessary watering and forms a basic foundation for smart agriculture and IoT-based irrigation systems.

15. Short Viva Points

• **What is the project?** An automatic irrigation system controlled by Arduino according to soil moisture.

• **Why is A0 used?** A0 is an analog input used to read the variable sensor signal.

• **What is the threshold?** 500 in the supplied program.

• **When does the pump start?** When sensorValue is less than 500.

• **When does the pump stop?** When sensorValue is 500 or higher.

• **Why is a transistor used?** It acts as a switching stage so the Arduino can control the motor safely.

• **Why is a diode connected across the motor?** To suppress the voltage spike produced when the motor is switched OFF.